中交海上风电智能

主线：PDF 文档解析

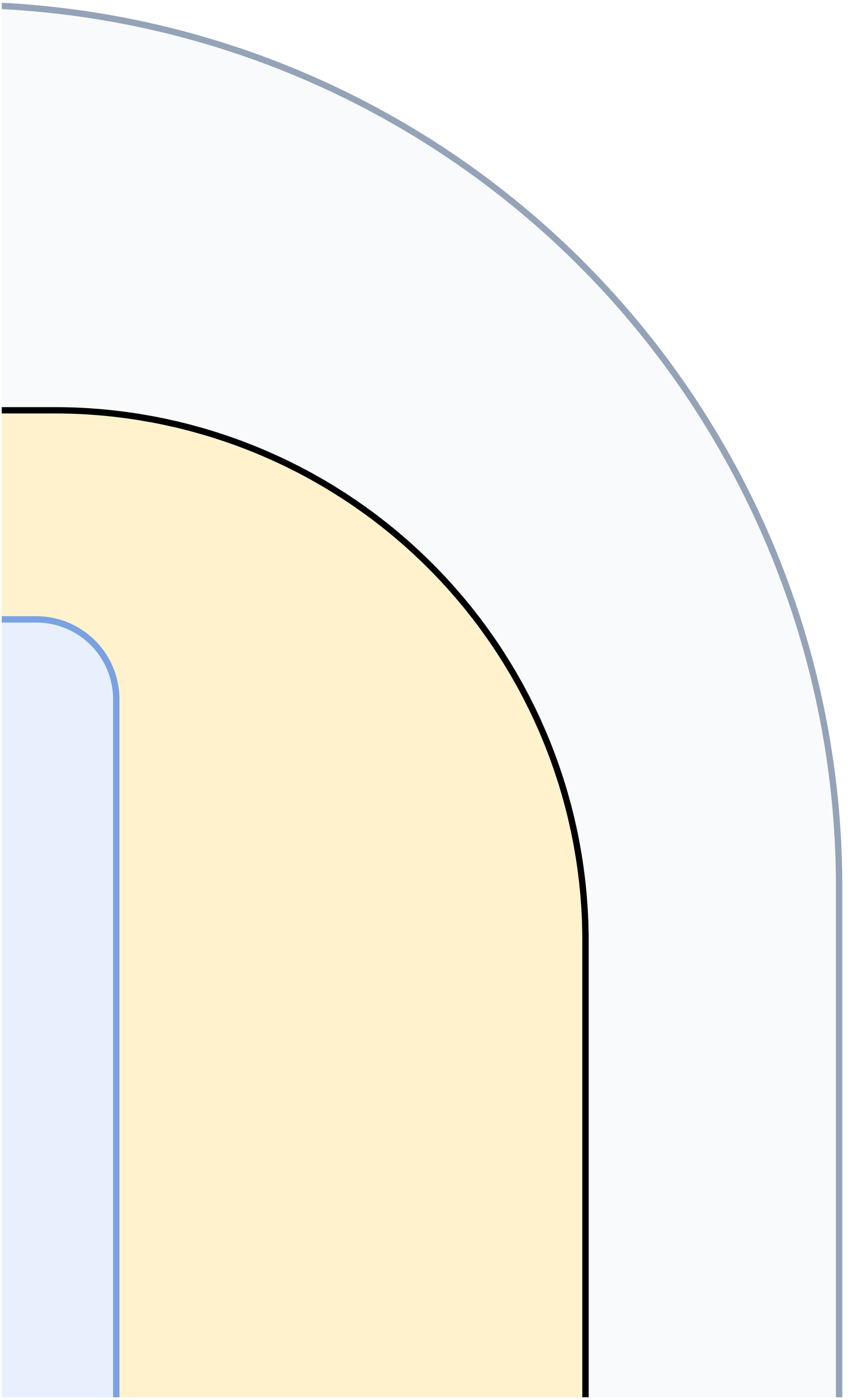
能体平台架构图 R

析、切分、编码、Milvus 入库

RAG 入库与检索流

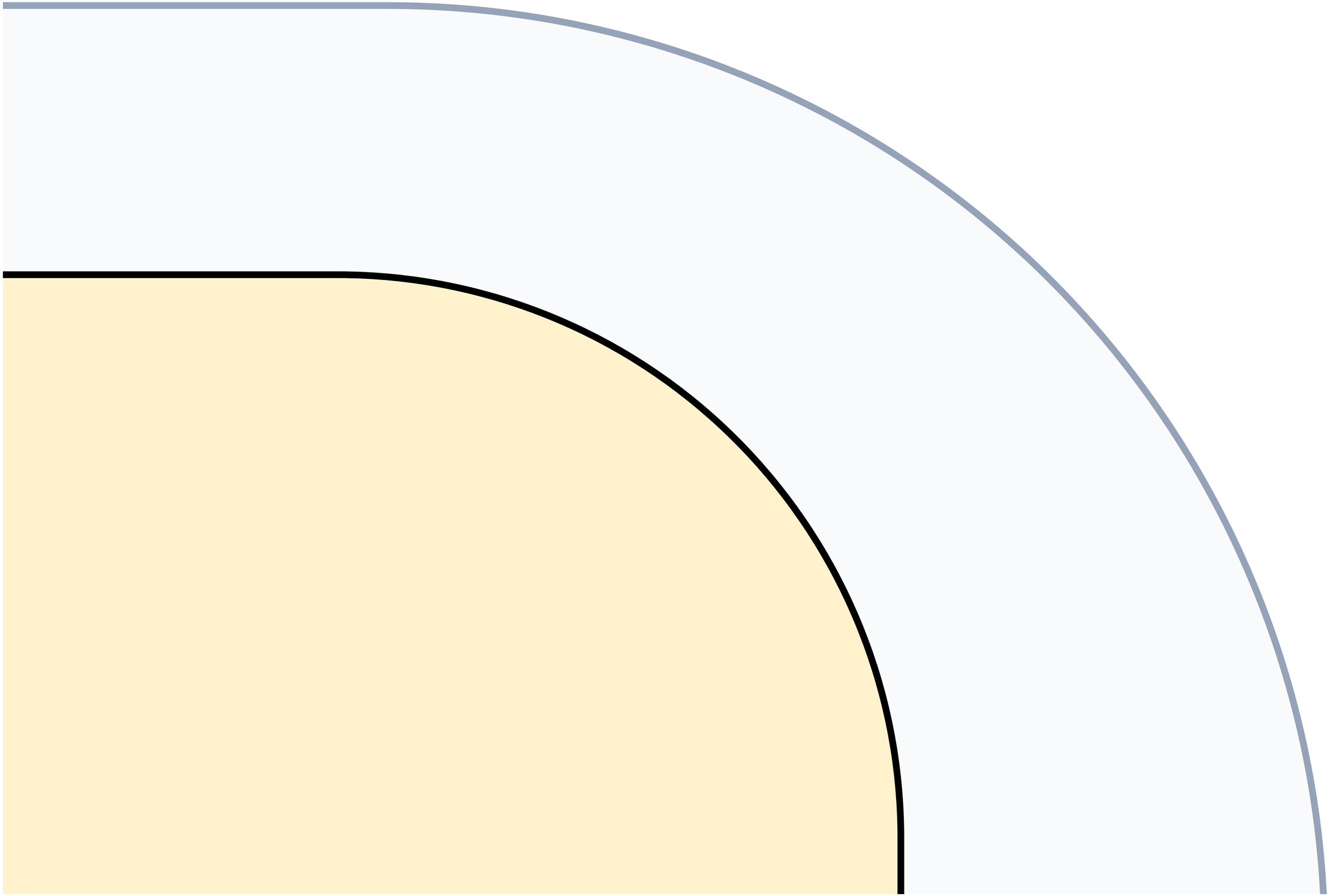
，以及查询侧的混合检索流程

流程图

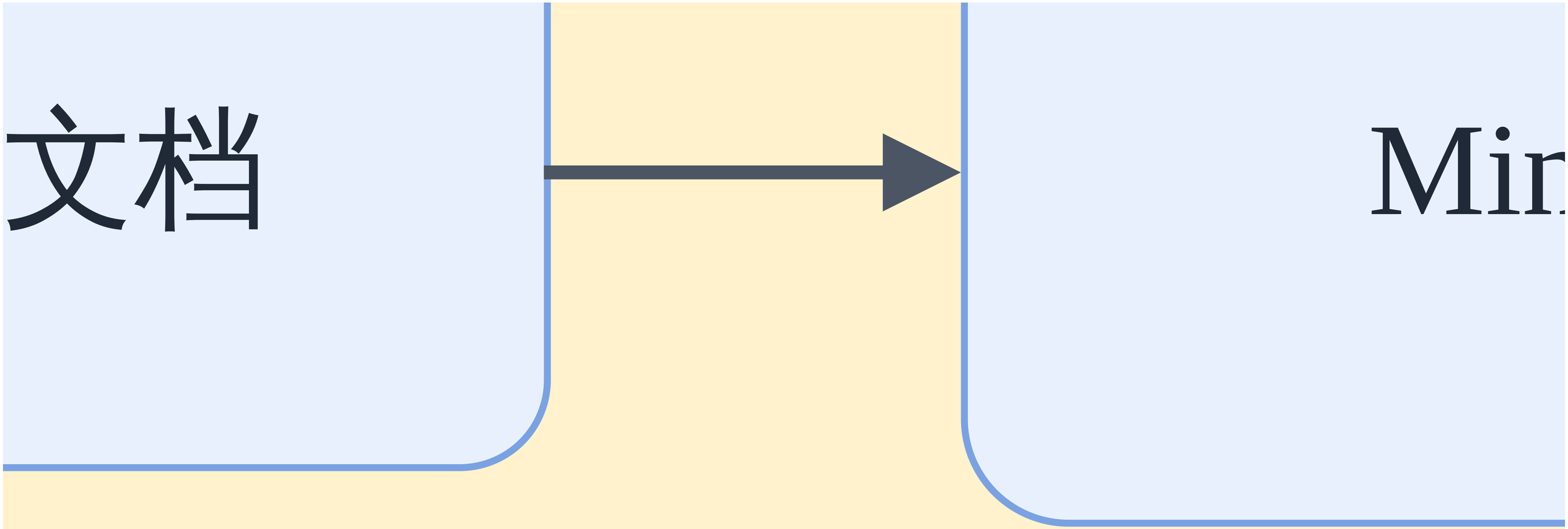


文档切分

分入库

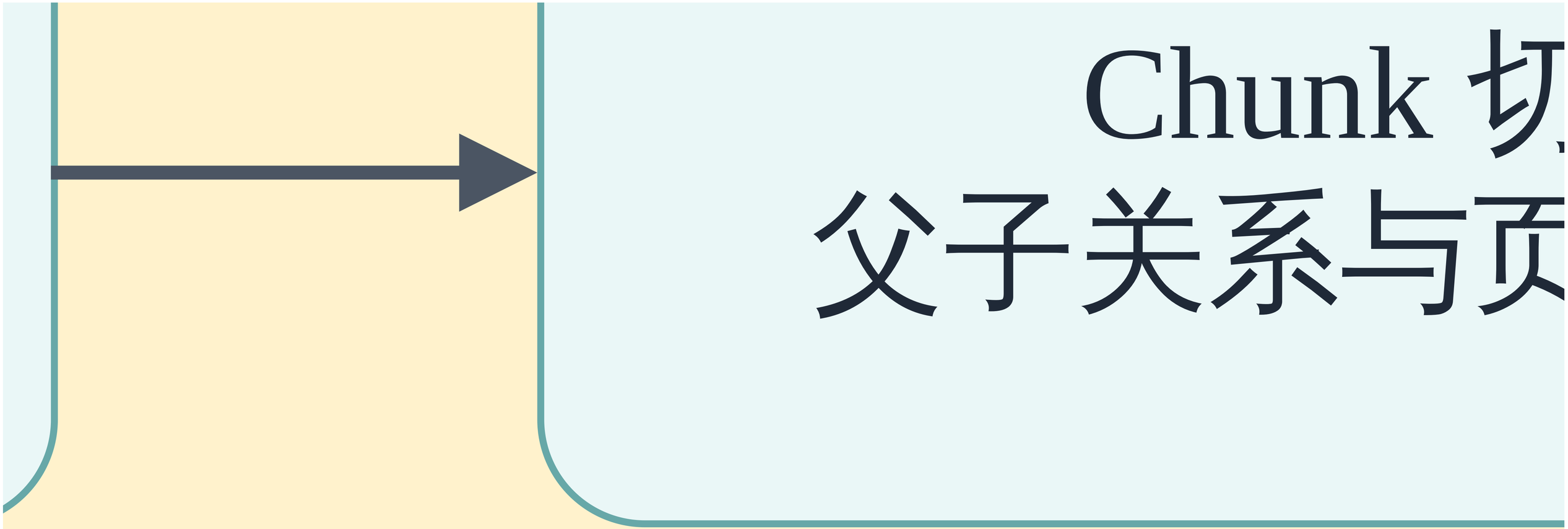


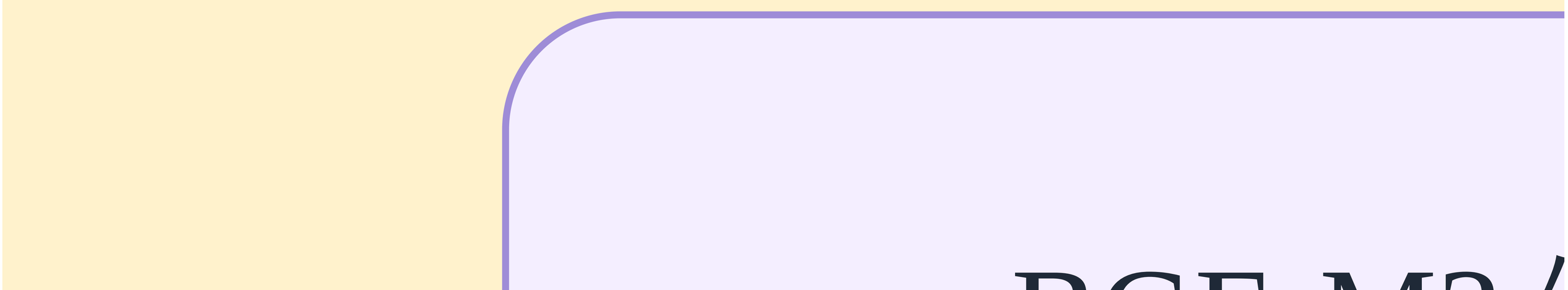
原始 PDF

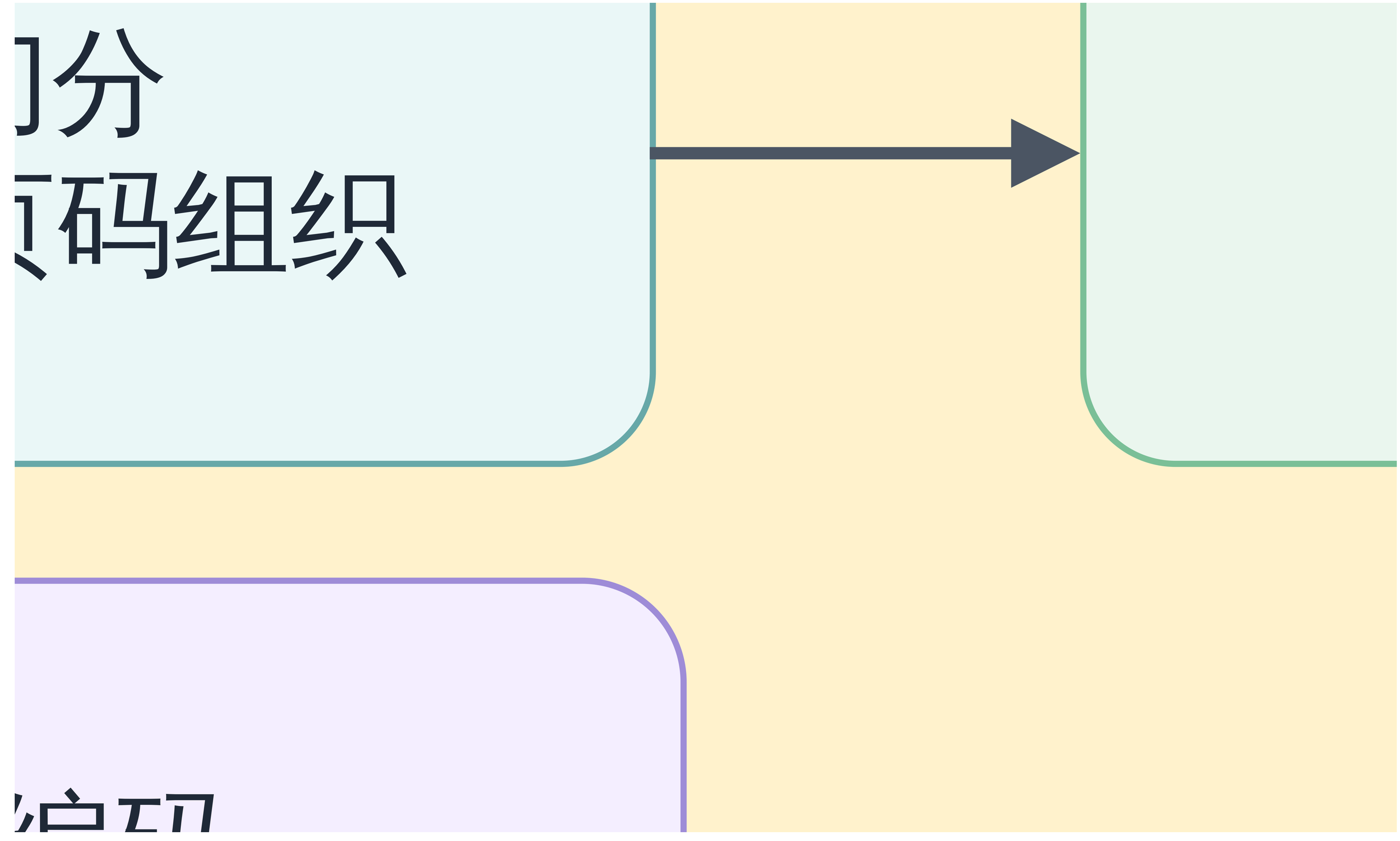

文档
Mir

nerU 解析

文本 / 表格 / 图片
结构化抽取


Chunk 切
父子关系与页


BGE-M3 4


分
码组织

LangChain JSONL 文档对象落盘

Milvus 4

知识库

BGE-M3 ⽣
成 dense + sp

编码

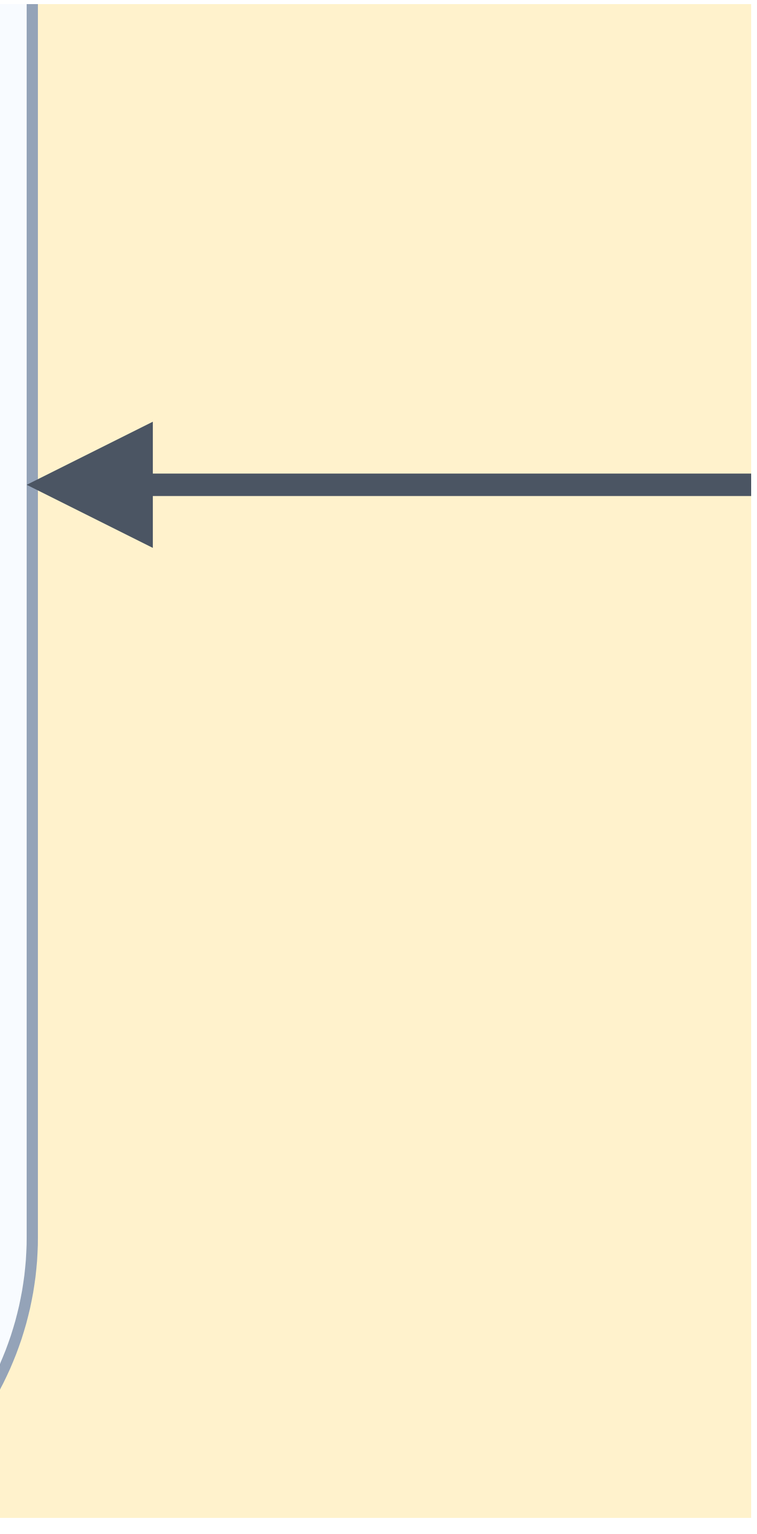
parse 表征

metadata 拆分

轻量 metadata / 完整 metadata

milvus
dense_
bge_spars
bm25_spars
轻量 元

解数据
_vector
se_vector
se_vector 等
元数据

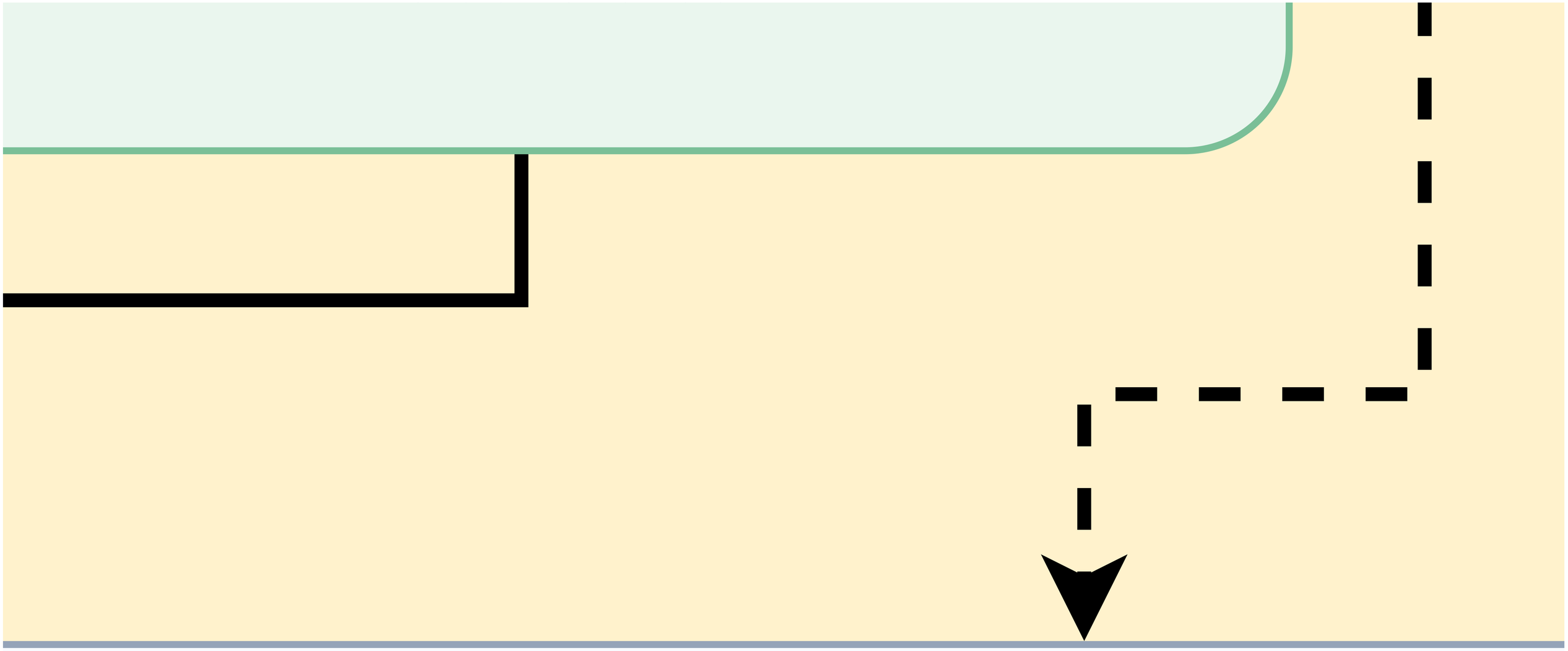


实体写入

Milvus BM25 Fu
BM25 spa

uction生成

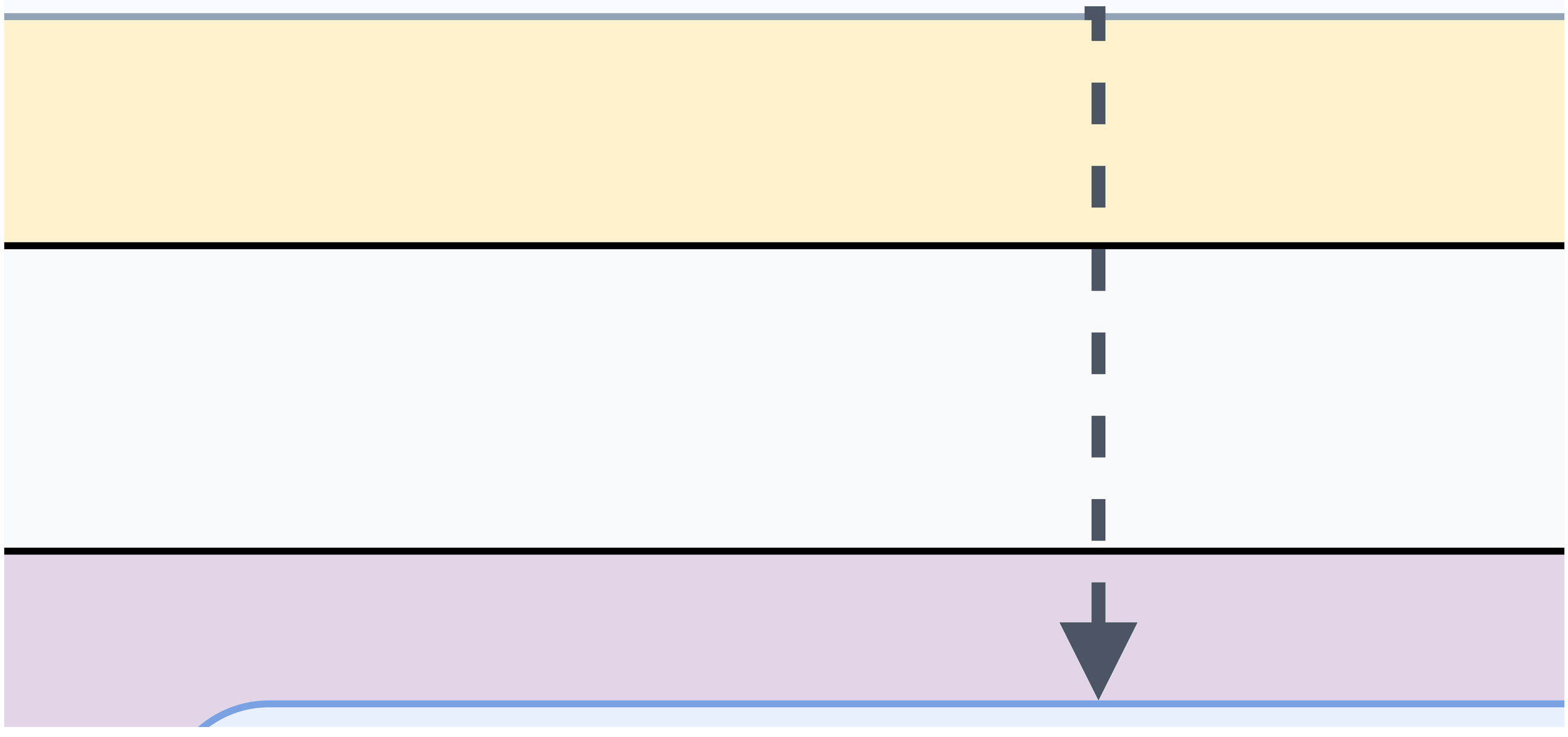
arse



完整 元数据 外置存

儲

用于完整的引用、图片、



表格回溯

查询与

检索

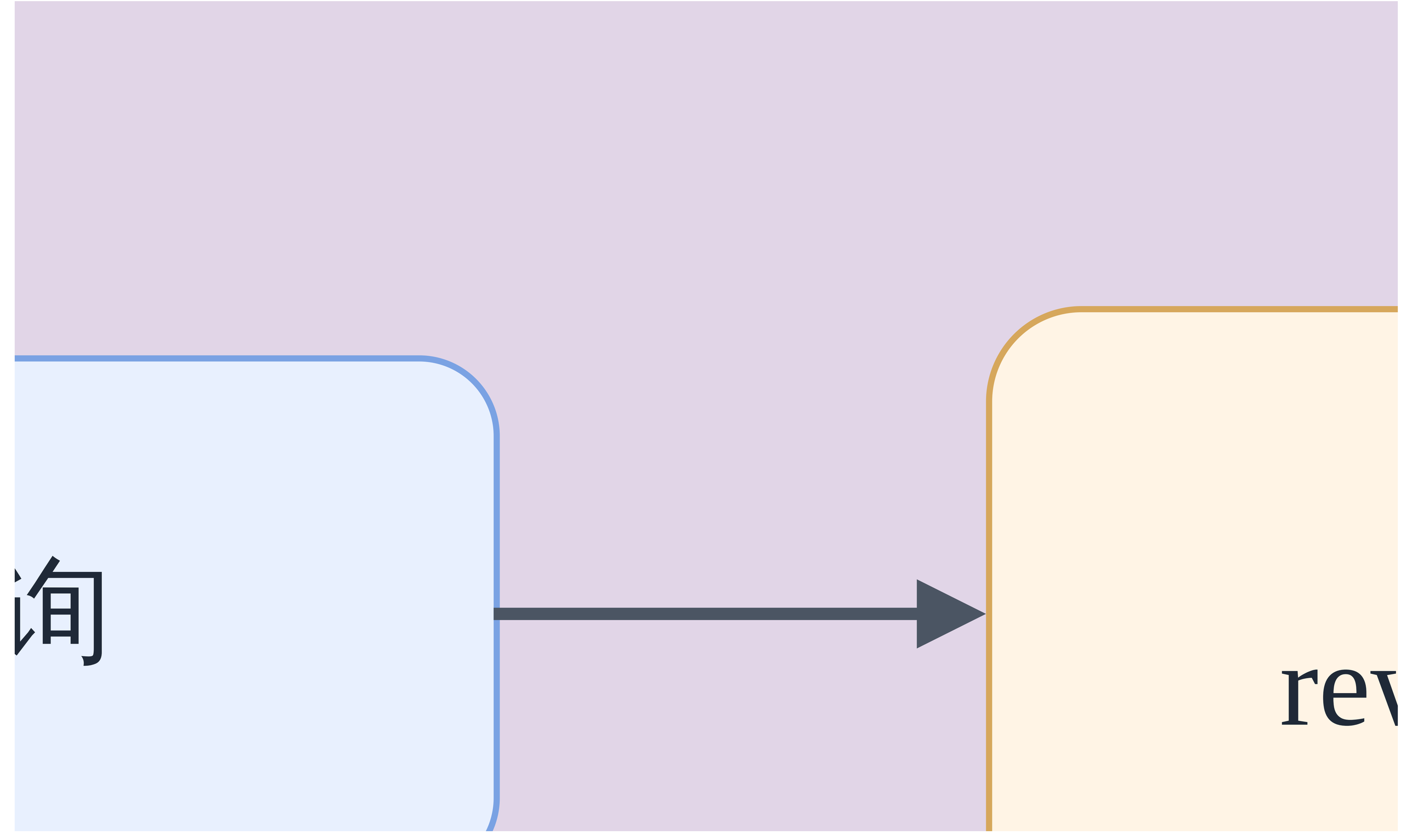
最终 Top-K 上下文
用于后续回答生成

文

rerank

候选重排

用户查


询
rev

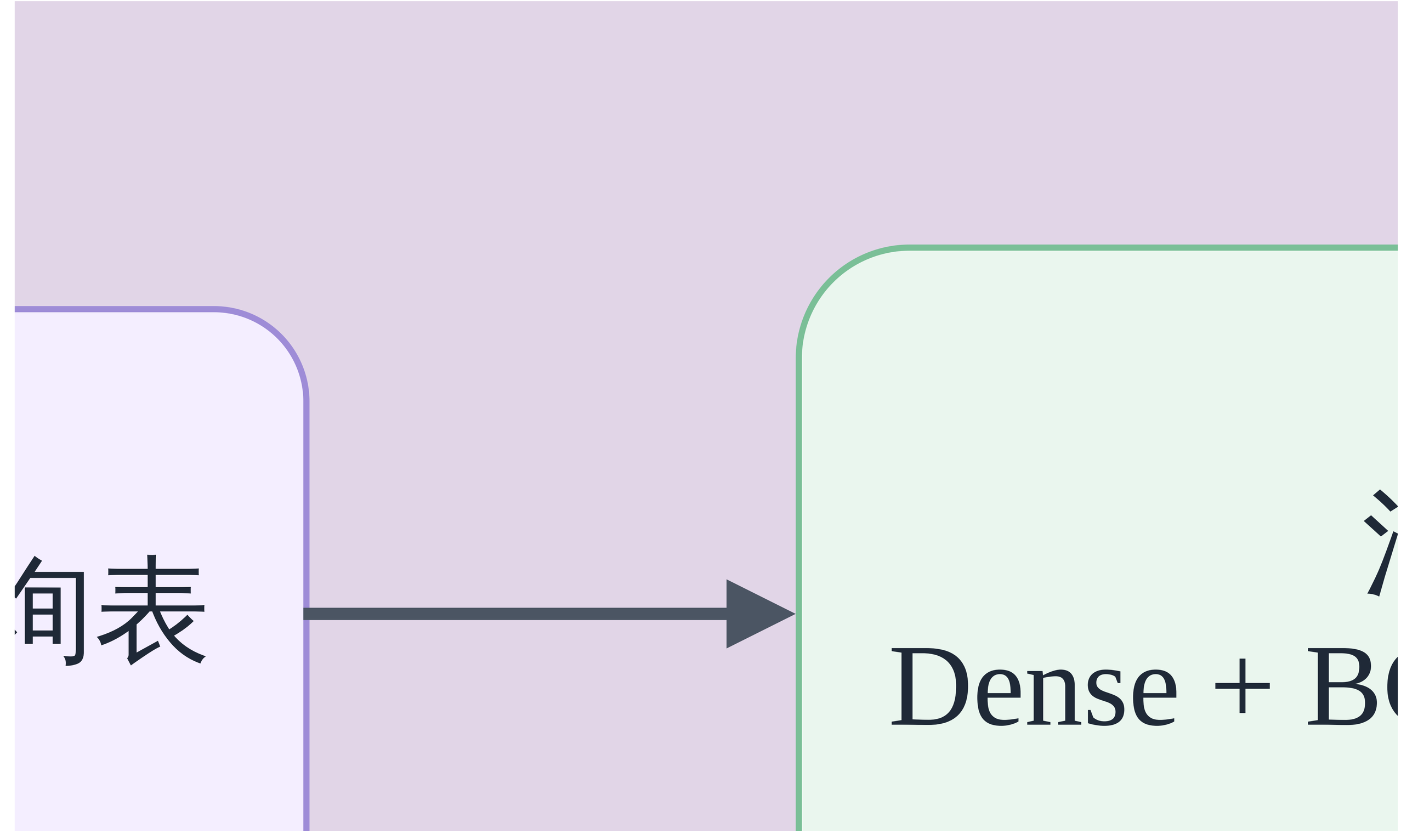
查询增强

write / 领域扩展

查询编码

生成 dense / sparse 查询

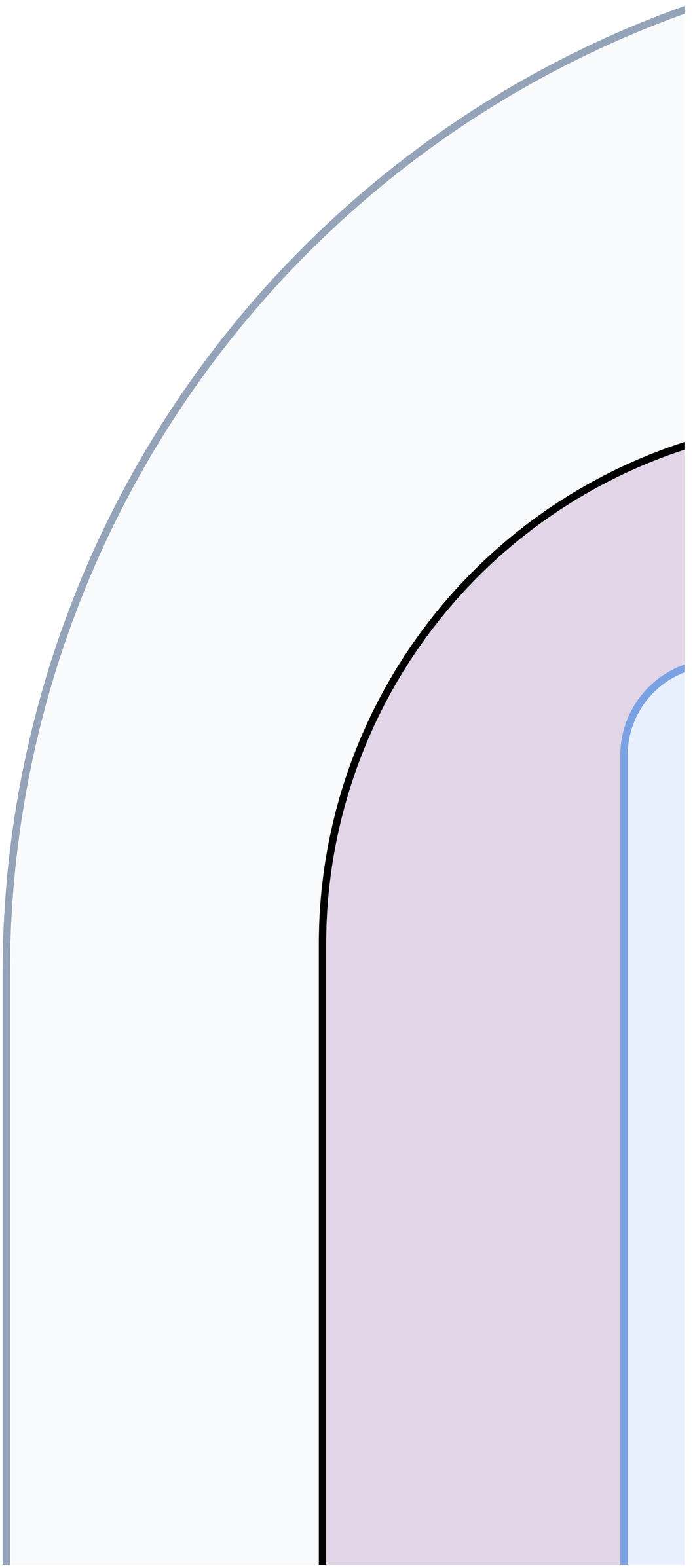
示

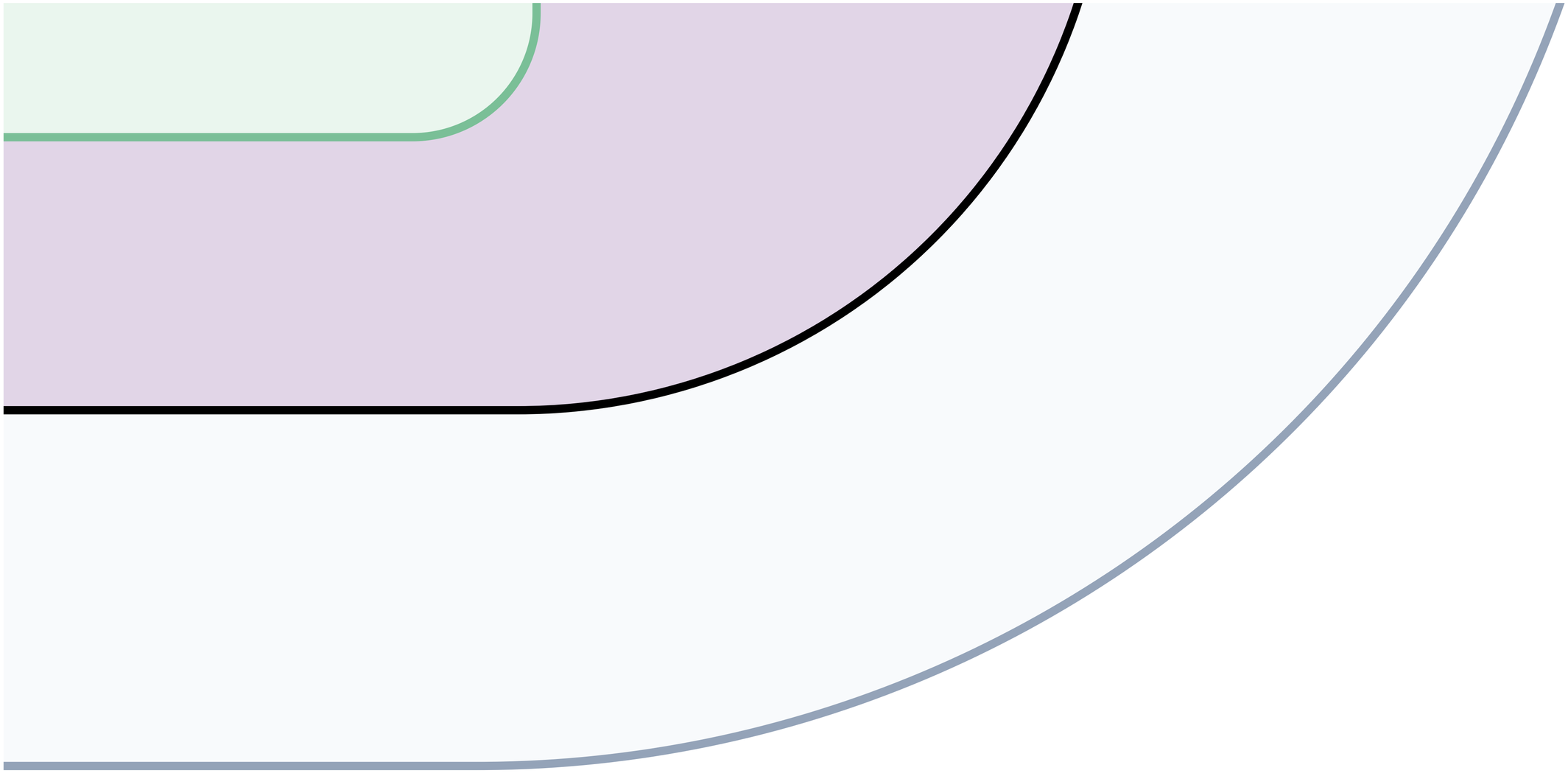

询表
Dense + B

混合检索

GE Sparse + BM25

RRF 融合 / 去重
文档聚合





说明：流程图中部分内容存在差异

，具体信息以代码仓库及实际应用为准